

DSPO: STABLE AND EFFICIENT POLICY OPTIMIZATION FOR AGENTIC SEARCH AND REASONING

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ABSTRACT

Endowing LLMs with the capacity to interrogate external knowledge sources is increasingly indispensable for complex, open-world tasks whose solution requires information that cannot be assumed to reside in parametric memory. Existing approaches either depend on prompting to elicit latent agentic behaviors, thereby leaving optimization largely implicit, or encounter performance ceilings and training collapse when RL is applied to complex interactive settings, so that the models’ agentic potential remains only partially realized. To address these limitations, we introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, an RL algorithm that promotes robust agent training by coupling sequence-level policy optimization with dynamic sample filtering. Our model is trained purely through RL to interleave multi-turn search and reasoning, thereby eliminating the need for supervised demonstration data. Across multiple QA benchmarks, our 7B model improves over a comparable previous work by **34.1%** and, while maintaining exceptional training stability, even exceeds the 14B model from previous work on complex multihop QA such as HotpotQA by nearly **9% relative**.

1 INTRODUCTION

Large Language Models (LLMs) (Brown et al., 2020; Touvron et al., 2023; Zhao et al., 2023) have exhibited exceptional competence across a broad range of specialized domains, including mathematical reasoning (Shao et al., 2024; Trinh et al., 2024; Yu et al., 2025), code generation (Zheng et al., 2023; Yang et al., 2024), and creative writing (Chakrabarty et al., 2024; Marco et al., 2024). Yet a fundamental constraint remains: their knowledge is intrinsically static, being bounded by the data distribution on which pretraining was conducted. A prominent strategy for mitigating this knowledge cutoff is therefore to augment LLMs with search capabilities, thereby transforming them into agents that can actively query external knowledge sources when their internal parametric memory is insufficient (Jin et al., 2025b). This capability constitutes a canonical instance of tool-calling (Schick et al., 2023), in which the model must learn to coordinate its own generation with interactions involving an external search tool. Acquiring such competence requires the induction of a complex, multi-step policy, which naturally casts agentic search as a sequential decision-making problem well suited to Reinforcement Learning (RL).

In contrast to Supervised Fine-Tuning (SFT), which depends on expensive static demonstrations and provides little opportunity for exploration, RL offers a mechanism through which LLMs can acquire effective policies via trial-and-error interaction (Chu et al., 2025). Within this setting, value-free methods such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024) have become influential because they simplify training while reducing memory overhead. Nevertheless, although GRPO is effective in more constrained regimes, its application to open-ended interactive search exposes substantial instabilities (Jin et al., 2025b; Yu et al., 2025; Cui et al., 2025; Liu et al., 2025). This fragility arises from two coupled deficiencies. First, as identified by Zheng et al. (2025), GRPO’s token-level objective is structurally mismatched with sequence-level rewards, producing high-variance gradients that destabilize policy updates. Second, because search tasks typically provide sparse rewards, rollout groups frequently exhibit homogeneous outcomes, such as all-success or all-failure samples; in such cases, the advantage signal collapses or becomes uninformative, thereby severely degrading sample efficiency (Yu et al., 2025; Liu et al., 2025).

To confront these intertwined problems of instability and inefficient learning, we introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**. The algorithm synthesizes and refines principles from recent policy-optimization work: it adopts the sequence-level optimization perspective of GSPO (Zheng et al., 2025), so that the unit of optimization coincides with the unit at which reward is assigned, and it thereby aligns the surrogate objective with the trajectory-level signal that governs long-horizon reasoning. In addition, DSPO incorporates a dynamic outcome-based filtering mechanism inspired by DAPO (Yu et al., 2025), constructing training batches from rollout groups that contain

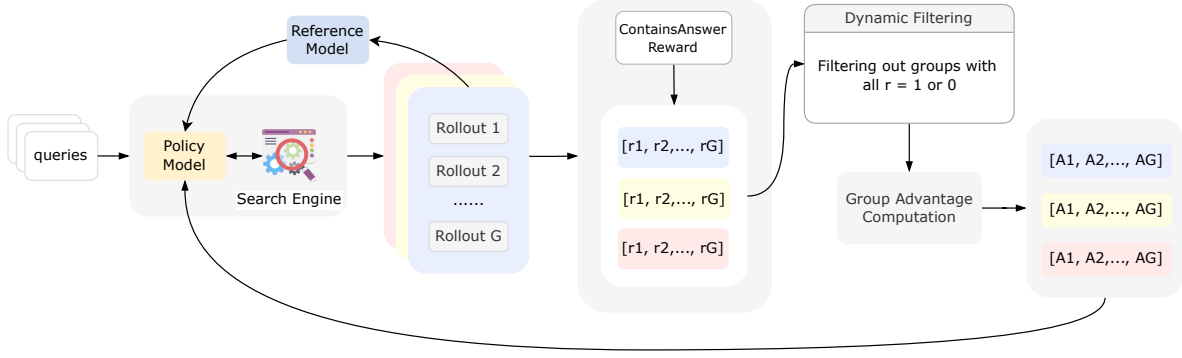


Figure 1: Overview of the DSPO training loop. For a given query, the policy model generates a group of G trajectories through interaction with the search environment, after which each trajectory receives a sparse terminal reward. The **dynamic filter** removes groups with homogeneous outcomes and continues sampling until the batch is filled, thereby ensuring that every training batch supplies an effective advantage signal. The resulting advantages are then used to update the policy model through the sequence-level objective.

both successful and unsuccessful outcomes for each prompt and consequently ensuring that the advantage signal $\hat{A}_i$ remains informative and stable. By integrating these two components into a unified framework, DSPO yields a stable, high-performance algorithm for complex multi-turn search and reasoning. Empirically, our model achieves a **34.1% relative improvement** over a leading 7B baseline (Jin et al., 2025b) and even surpasses its 14B counterpart (Jin et al., 2025a) on complex multi-hop benchmarks such as HotpotQA, where it obtains a **nearly 9% relative gain** (0.613 vs. 0.563).

In summary, the paper makes the following contributions:

- We propose DSPO, an RL algorithm that addresses the central instability and sample-inefficiency pathologies that arise when training agentic search models. The method unifies two complementary principles within a single coherent framework: **sequence-level optimization**, which supports robust policy updates, and **dynamic outcome-based filtering**, which densifies the otherwise sparse learning signal.
- We substantiate DSPO’s performance gains through rigorous benchmarking. Our 7B model achieves a **34.1% relative improvement** over a comparable 7B baseline and, more strikingly, exceeds its 14B counterpart on complex multi-hop QA, yielding a nearly **9% relative gain** on HotpotQA (0.613 vs. 0.563).
- We provide extensive empirical evidence that DSPO induces markedly more stable training trajectories. Crucially, these results are obtained using only a basic BM25 retriever, which isolates the observed performance gains to the robustness of the optimization algorithm rather than to retrieval-system sophistication.

2 RELATED WORK

2.1 RL FOR LLMs

The methodological landscape of RL for LLMs has developed rapidly, progressing from foundational Reinforcement Learning from Human Feedback (RLHF) procedures that rely on PPO and explicit reward models (Ouyang et al., 2022; Christiano et al., 2017; Schulman et al., 2017) toward simpler direct-optimization frameworks such as DPO (Rafailov et al., 2023). A particularly consequential movement toward value-free optimization is represented by Group Relative Policy Optimization (GRPO) (Shao et al., 2024), which simplifies training by deriving learning signals from group-level reward statistics. However, because GRPO retains a token-level objective, it is susceptible to training instability (Liu et al., 2025; Cui et al., 2025), motivating several targeted refinements. GSPO addresses the reward-objective mismatch by adopting a sequence-level objective aligned with the unit of reward (Zheng et al., 2025), whereas DAPO mitigates inefficient learning under sparse rewards through a dynamic outcome-based sampling mechanism (Yu et al., 2025). Relatedly, GMPO stabilizes token-level optimization by using geometric-mean aggregation to reduce sensitivity to outliers (Zhao et al., 2025). Despite these advances, our experiments indicate that these algorithms can still encounter training collapse or performance bottlenecks. Building on this body of work, we propose an improved algorithm that synthesizes sequence-level optimization with dynamic filtering and filling, thereby addressing the distinctive difficulties of training autonomous search agents within a unified policy-optimization framework.

2.2 LLMs WITH AGENTIC RETRIEVAL

To mitigate the static-knowledge limitations of LLMs, retrieval-augmented generation (RAG) integrates external retrievers so that models can incorporate dynamically evolving information (Lewis et al., 2020; Gao et al., 2023). Classical RAG frameworks typically employ dense retrievers to obtain relevant documents, which are then concatenated into the LLM input before generation (Karpukhin et al., 2020); however, because these systems usually instantiate fixed pipelines, they offer limited autonomy in complex multi-turn settings. Recent work has therefore shifted toward agentic paradigms in which LLMs iteratively plan, search, and reason. Frameworks such as ReAct couple reasoning with action, enabling tool interaction for tasks including web navigation (Yao et al., 2023), while multi-agent systems such as AutoGen facilitate collaborative workflows (Wu et al., 2024). More recent innovations emphasize the integration of agentic RAG with RL, where retrieval is improved through decision-making rather than merely appended as preprocessing. Wu et al. (2025) introduce Agentic Reasoning, a framework that incorporates external tools to streamline LLM reasoning, and several RL-integrated approaches (Jin et al., 2025b; Chen et al., 2025; Song et al., 2025) train LLMs, using purely RL, to interleave reasoning and search. This end-to-end paradigm internalizes agentic capabilities and can reduce the engineering overhead associated with multi-agent frameworks. Nevertheless, such methods continue to face training instability and performance limitations in open-ended search domains. Our work directly targets these bottlenecks: DSPO supplies a robust and efficient training framework that stabilizes policy optimization and enables LLMs to acquire effective multi-turn search strategies.

3 METHODOLOGY

This section first formulates agentic search as an RL problem and reviews prior policy-optimization algorithms, with particular attention to the limitations that become salient in this setting. We then introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, specifying the mechanisms through which its core components improve both training stability and training efficiency. Finally, we present the integrated training algorithm.

3.1 PRELIMINARIES

Policy Gradient Methods for LLMs. RL-based training of LLMs commonly relies on policy-gradient methods such as PPO (Schulman et al., 2017), which has become a widely used algorithm for LLM alignment. PPO optimizes a policy π_θ by maximizing a clipped surrogate objective computed from samples generated by an earlier policy $\pi_{\theta_{\text{old}}}$. The token-averaged objective is given by:

$$J_{\text{PPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} \min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (1)$$

where $r_t(\theta) = \frac{\pi_\theta(y_t|x, y_{<t})}{\pi_{\theta_{\text{old}}}(y_t|x, y_{<t})}$ denotes the token-level importance ratio. PPO, however, requires a separately trained value model in order to estimate token-level advantages $\hat{A}_t$ via Generalized Advantage Estimation (GAE) (Schulman et al., 2015), a design choice that introduces substantial memory overhead and may itself become a source of optimization instability.

GRPO (Shao et al., 2024) was proposed to remove this dependence on a value model. For a given prompt x , GRPO samples a group of G responses $\{y_i\}_{i=1}^G$ and computes each response’s advantage by normalizing its reward with respect to the statistics of the group. Despite this group-relative construction, the optimization objective remains token-level, as in PPO:

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip}(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) - \beta D_{KL}(\pi_\theta || \pi_{\text{ref}}) \right], \quad (2)$$

where $r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})}$, and the advantage for every token $y_{i,t}$ in a response y_i is set to the same sequence-level value:

$$\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}, \quad (3)$$

Thus, every token in a response y_i is assigned the identical advantage $\hat{A}_i$, even though that quantity is derived from a reward defined only at the sequence level.

Agentic Search as a Markov Decision Process. We formalize the iterative procedure of agentic search and reasoning as a sequential decision-making problem, specifically a discrete-time, finite-horizon Markov Decision Process (MDP).

- **State (s_t):** At turn t , the state s_t encodes the complete interaction history, including the initial question q , all previously generated thoughts and search queries, and the retrieved evidence returned by the environment. Because retrieved evidence must be interpreted and integrated into subsequent decisions, the state expands over the trajectory, producing long-horizon dependencies that conventional token-level RL methods struggle to optimize.
- **Action (a_t):** An action is a complete textual segment generated by the policy π_θ , comprising free-form reasoning followed by a decision. The segment terminates either with a `</tool_call>` token, which triggers a search and returns results within `</tool_response>`, or with a `</answer>` token, which terminates the trajectory. This structured action space requires the agent to learn not only *what* to generate but also *when* to search, thereby introducing substantial variability in trajectory length.
- **Policy (π_θ):** The policy π_θ is the underlying LLM, which generates tokens autoregressively conditioned on the current state. Although the policy conditions on the entire interaction history, the optimization target is computed only over model-generated thoughts and actions, with retrieved environmental content masked out (Jin et al., 2025b). Optimizing this policy therefore requires credit assignment across long sequences in which many intermediate reasoning steps receive neither direct supervision nor intermediate reward, further motivating stable sequence-level optimization.
- **Trajectory (τ):** A trajectory $\tau = (s_1, a_1, \dots, s_T, a_T)$ records all reasoning and tool interactions undertaken for a given question, including both model-generated actions and search results returned by the environment. Since the final reward is assigned to the trajectory as a whole, the optimization problem is intrinsically sequence-level: early decisions may shape evidence acquisition, reasoning context, and ultimately answer correctness.
- **Reward ($R(\tau)$):** We employ a sparse terminal reward, under which a trajectory receives $R = 1$ if its final answer contains the ground-truth text and $R = 0$ otherwise:

$$R(\tau) = \begin{cases} 1 & \text{if } a_{\text{gold}} \subseteq a_{\text{pred}}, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

RL with a Search Engine. Following Jin et al. (2025b), we explicitly treat the search engine, denoted by $\mathcal{S}$, as a component of the environment. The policy LLM π_θ learns to generate trajectories by interleaving reasoning steps with calls to $\mathcal{S}$. The resulting optimization problem is to identify a policy that maximizes expected reward while being regularized by a KL divergence term that discourages large deviations from a reference policy π_{ref} :

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x;\mathcal{S})} [R(x, y)] - \beta D_{\text{KL}} [\pi_\theta(y|x;\mathcal{S}) || \pi_{\text{ref}}(y|x;\mathcal{S})]. \quad (5)$$

Here, $y \sim \pi_\theta(\cdot|x;\mathcal{S})$ indicates that the trajectory y is produced through a multi-step interaction involving both the policy’s token generation and the information returned by the search engine $\mathcal{S}$.

Motivation for DSPO. Although frameworks such as Search-R1 (Jin et al., 2025b) have productively cast agentic search as an RL problem, the application of conventional algorithms such as PPO or GRPO remains difficult. The open-endedness of the search environment amplifies the instability of token-level optimization, principally because the unit of reward assignment is the sequence whereas the unit of optimization is the token (Zheng et al., 2025). This discrepancy yields high-variance gradient estimates that accumulate across long trajectories and can culminate in policy collapse. Moreover, the sparse binary reward signal causes many training batches to contain only successful or only unsuccessful trajectories, producing abnormal or null advantages and therefore little usable learning signal, which sharply reduces sample efficiency (Yu et al., 2025; Liu et al., 2025). DSPO is designed to counteract these two failure modes directly.

3.2 DYNAMIC-FILTER SEQUENCE-LEVEL POLICY OPTIMIZATION

DSPO introduces two principal innovations relative to prior methods: (1) policy optimization is performed at the sequence level, so that the training objective is aligned with the trajectory-level reward structure; and (2) a dynamic

filtering mechanism ensures that every retained training batch provides a high-quality, non-zero learning signal. The complete training procedure, which integrates these components, is illustrated in Figure 1.

3.2.1 SEQUENCE-LEVEL POLICY OPTIMIZATION FOR ENHANCED STABILITY

Motivated by GSPO (Zheng et al., 2025), we replace the unstable token-level importance ratio with a theoretically better aligned sequence-level counterpart. For a response y_i , the sequence-level importance ratio $s_i(\theta)$ is defined as the geometric mean of its token-level ratios:

$$s_i(\theta) = \left(\frac{\pi_\theta(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)} \right)^{\frac{1}{|y_i|}} = \exp \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})} \right). \quad (6)$$

This length normalization is essential because it reduces variance and keeps $s_i(\theta)$ within a consistent numerical range across responses of different lengths, a property that is crucial for stable clipping.

Gradient Analysis. The following gradient comparison clarifies why DSPO improves stability. In the unclipped token-level GRPO objective, each token’s log-probability gradient is scaled by a potentially noisy, token-specific weight $r_{i,t}(\theta)$. By contrast, the gradient of our sequence-level objective scales the log-probability gradient aggregated over the entire sequence by a single, more stable sequence-level weight $s_i(\theta)$:

$$\nabla_\theta J_{\text{GRPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot \sum_{t=1}^{|y_i|} r_{i,t}(\theta) \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (7)$$

$$\nabla_\theta J_{\text{DSPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot s_i(\theta) \cdot \sum_{t=1}^{|y_i|} \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (8)$$

By applying one holistic correction factor to the whole trajectory, DSPO avoids the accumulation of token-level noise that undermines prior methods and thereby yields substantially more stable training. In parallel, the dynamic filtering mechanism preserves a well-defined advantage signal $\hat{A}_i$ by constructing training batches from rollout groups that contain both successes and failures, preventing sample waste in sparse-reward environments.

3.2.2 DYNAMIC OUTCOME-BASED FILTERING FOR EFFICIENT LEARNING

The sparse binary structure of our reward function creates a significant obstacle for group-based advantage estimation. When all G responses in a group are correct ($R = 1$) or all are incorrect ($R = 0$), the normalized advantage $\hat{A}_i$ becomes either zero or undefined, so the corresponding batch contributes no useful gradient signal while still consuming computation.

To address this inefficiency, DSPO incorporates a dynamic filtering mechanism inspired by DAPO (Yu et al., 2025). During sampling, we retain only trajectory groups that contain a mixture of successful and unsuccessful outcomes. A group $\{y_i\}_{i=1}^G$ is admitted to training only when its rewards $\{R_i\}_{i=1}^G$ satisfy:

$$0 < \sum_{i=1}^G R_i < G. \quad (9)$$

This condition ensures non-zero reward variance within every retained training group and therefore guarantees a meaningful advantage signal. Through this dynamic selection, each policy update is computed from a curated high-quality sample set, effectively converting a sparse-reward regime into a denser and more computationally efficient learning process.

3.3 THE DSPO OBJECTIVE AND TRAINING ALGORITHM

Integrating these components yields the final DSPO objective. For each valid group drawn from the filtered sample space $\mathcal{D}_{\text{filtered}}$, we compute the advantage $\hat{A}_i$ through group-relative normalization:

$$\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R) + \delta}, \quad (10)$$

Algorithm 1 Dynamic-filter Sequence-level Policy Optimization (DSPO)

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1: Input: Initial policy  $\pi_{\theta_0}$ , fixed reference policy  $\pi_{\text{ref}}$ , prompt dataset  $\mathcal{D}$ , group size  $G$ , batch size  $B$ , search tool  $\mathcal{R}$ .
2: Initialize policy  $\pi_{\theta} \leftarrow \pi_{\theta_0}$ .
3: for each training step do
4:    $\pi_{\theta_{\text{old}}} \leftarrow \pi_{\theta}$ .
5:   Initialize training buffer  $\mathcal{B} \leftarrow \emptyset$ .
6:   while  $|\mathcal{B}| < B$  do
7:     Sample a prompt  $x \sim \mathcal{D}$ .
8:     Generate a group of  $G$  trajectories  $\{y_i\}_{i=1}^G$  using  $\pi_{\theta_{\text{old}}}$  and the search tool  $\mathcal{R}$ .
9:     Compute terminal rewards  $\{R_i\}_{i=1}^G = \{\text{ContainsAnswer}(y_i, y_{\text{gold}})\}_{i=1}^G$ .
10:    if  $0 < \sum_{i=1}^G R_i < G$  then ▷ Dynamic outcome-based filtering
11:      Add  $(x, \{y_i\}_{i=1}^G, \{R_i\}_{i=1}^G)$  to  $\mathcal{B}$ .
12:    end if
13:  end while
14:  for each  $(x, \{y_i\}, \{R_i\})$  in  $\mathcal{B}$  do
15:    Compute advantages  $\{\hat{A}_i\}_{i=1}^G$  via group normalization of  $\{R_i\}$ .
16:    Compute sequence-level importance ratios  $\{s_i(\theta)\}_{i=1}^G$  using Eq. 6.
17:    Compute the DSPO loss for the group using Eq. 11, applying masks to retrieved tokens.
18:  end for
19:  Update policy parameters  $\theta$  by taking a gradient step on the total loss from  $\mathcal{B}$ .
20: end for

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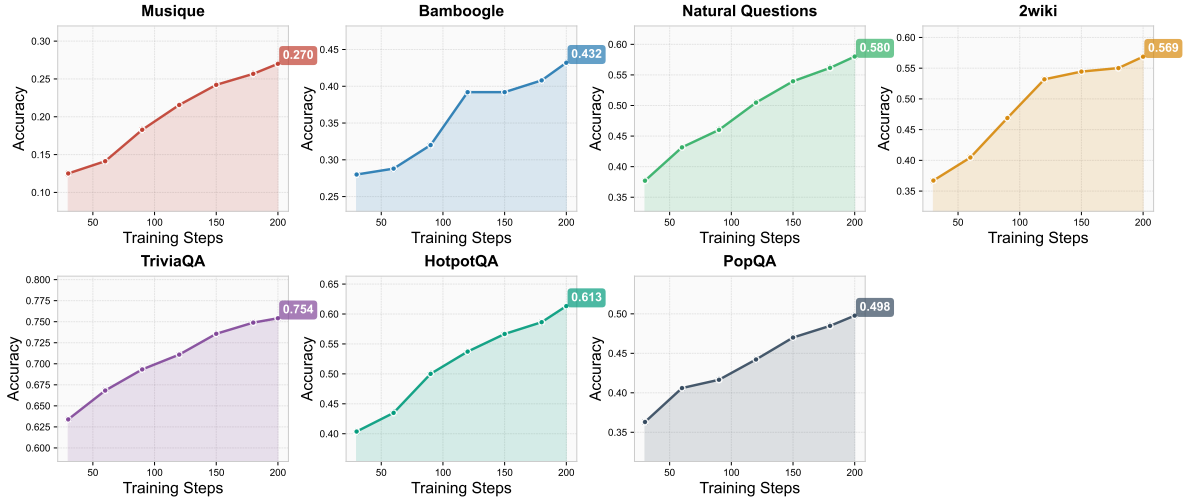


Figure 2: **Validation performance of DSPO across seven benchmarks during training.** The steady, monotonic improvement in accuracy indicates that DSPO’s reward gains translate into enhanced generalization, rather than merely optimizing the training signal, and that the method learns a robust search-and-reasoning policy.

where δ is a small numerical-stability constant. The policy π_{θ} is then updated by maximizing:

$$J_{\text{DSPO}}(\theta) = \mathbb{E}_{(x, \{y_i\}) \in \mathcal{D}_{\text{filtered}}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right], \quad (11)$$

where s_i is the sequence-level importance ratio defined as the geometric mean of the token-level ratios. During likelihood calculation, we additionally apply loss masking to all tokens retrieved from the search tool, following Jin et al. (2025b); this masking ensures that the model learns to use external knowledge for reasoning rather than merely reproducing retrieved text. The complete training process is detailed in Algorithm 1.

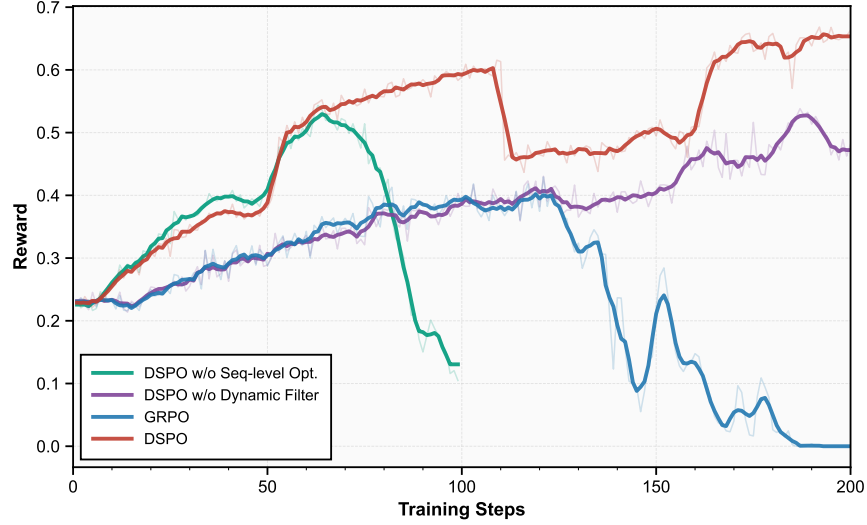


Figure 3: **Training reward dynamics of DSPO and its ablations.** The learning curves show that DSPO (red) achieves stable, monotonic improvement, whereas the token-level variants (green, blue) undergo catastrophic policy collapse and the sequence-level variant without the proposed filter (purple) stabilizes only at a suboptimal plateau.

4 EXPERIMENTS

This section presents a series of experiments designed to empirically validate the effectiveness of Dynamic-filter Sequence-level Policy Optimization (DSPO). The evaluation is organized around three objectives: demonstrating that DSPO achieves strong performance on challenging question-answering benchmarks; establishing that it substantially improves training stability by avoiding the catastrophic collapse observed in baseline methods; and verifying, through ablation, that its performance gains arise from the synergistic interaction of its core components.

4.1 EXPERIMENTAL SETUP

Following Search-R1 (Jin et al., 2025b), and as shown in Table 1, we use a prompt template that specifies the model’s permissible actions during search, including reasoning within `<think>`, tool invocation through `<tool_call>`, and final response generation within `<answer>`.

Prompt Template. Answer the given question. Whenever new information is obtained, first conduct reasoning inside `<think>` and `</think>`. After reasoning, if additional knowledge is required, invoke the search engine with `<tool_call>` query `</tool_call>`; the top retrieved results will be returned between `<tool_response>` and `</tool_response>`. You may search repeatedly as needed. Once no further external knowledge is necessary, provide the final answer directly inside `<answer>` and `</answer>`, without detailed explanation. For example, `<answer>` Beijing `</answer>`. **Question:** ...

Table 1: Prompt template used to structure reasoning, search invocation, and final-answer generation in our experiments.

Benchmarks and Baselines. To ensure a rigorous and comparable evaluation, our experimental design follows the established Search-R1 protocol (Jin et al., 2025b). We train the model on a composite dataset comprising the training splits of Natural Questions (NQ) (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018). We then evaluate generalization on the test sets of seven diverse QA benchmarks: NQ, TriviaQA (Joshi et al., 2017), PopQA (Mallen et al., 2022), HotpotQA, 2WikiMultiHopQA (Ho et al., 2020), Musique (Trivedi et al., 2022), and Bamboogle (Press et al., 2022).

The comparison suite contains both strong external baselines and diagnostic internal ablations. The external baselines are Qwen2.5-7B and 14B models trained with PPO and GRPO under the Search-R1 framework (Jin et al., 2025b;a).

To decompose the contribution of each DSPO component, we further include two internal ablations: (1) **DSPO w/o dynamic filter**, which is equivalent to GSPO (Zheng et al., 2025), and (2) **DSPO w/o sequence-level opt.**, which reverts to a strong token-level policy, DAPO (Yu et al., 2025).

For implementation, Qwen2.5-7B-Instruct serves as the initial checkpoint for all training runs. The experiments are implemented on the VeRL framework (Sheng et al., 2025), for which we adapt the provided Search-R1-like example code and scripts to match our methodology. DSPO is benchmarked against a comprehensive set of baselines, including PPO and GRPO methods from the Search-R1 framework (Jin et al., 2025b;a). Because our work uses a modified reward function, we retrain these models under the same experimental conditions rather than relying solely on previously reported numbers, and the resulting retrained models constitute our primary external benchmarks.

Implementation and Evaluation. To isolate the contribution of the optimization algorithm, all RL experiments deliberately use a standard BM25 retriever. This controlled setting ensures that observed performance improvements can be attributed to the learned policy rather than to improvements in retrieval infrastructure. Across all methods, models are trained with a sparse binary reward derived from substring Exact Match (subEM) of the final answer, and subEM is used as the primary evaluation metric.

4.2 MAIN RESULTS AND ABLATION STUDY

To provide a holistic assessment of the algorithm’s effectiveness, Table 2 reports a comprehensive comparison across benchmarks. Because DSPO’s components are designed to operate synergistically, we present the main results together with the ablation study, allowing the full method to be evaluated simultaneously against external state-of-the-art baselines and against variants in which individual components are removed. This organization provides a direct view of both DSPO’s overall advantage and the indispensability of its constituent mechanisms.

Table 2: Comprehensive comparison of DSPO, external baselines, and ablation variants across seven QA benchmarks. Baselines include Search-R1 models (7B & 14B) trained with GRPO and PPO (Jin et al., 2025b;a); ablations remove the dynamic filter or sequence-level optimization components. Original EM scores from Search-R1 are reported in parentheses, while consistency of evaluation is maintained by retraining and evaluating the relevant models under our adjusted rewards. Best results are in **bold**; second-best are underlined.

Dataset	Search-R1				DSPO & Ablations (Ours, 7B)		
	GRPO (7B)	PPO (7B)	GRPO (14B)	PPO (14B)	w/o dyn. filter	w/o seq-lvl opt.	DSPO
NQ	0.423 (0.429)	(0.393)	<u>0.535</u> (0.482)	(0.424)	0.363	0.470	0.580
TriviaQA	0.658 (0.623)	(0.610)	0.760 (0.667)	(0.660)	0.515	0.695	<u>0.754</u>
PopQA	0.395 (0.427)	(0.397)	<u>0.477</u> (0.434)	(0.442)	0.277	0.430	0.498
HotpotQA	0.401 (0.386)	(0.370)	<u>0.563</u> (0.429)	(0.436)	0.330	0.438	0.613
2WikiMultiHopQA	0.357 (0.414)	(0.346)	0.611 (0.424)	(0.379)	0.285	0.398	<u>0.569</u>
Musique	0.122 (0.162)	(0.146)	<u>0.260</u> (0.191)	(0.210)	0.105	0.133	0.270
Bamboogle	0.280 (0.400)	(0.368)	0.504 (0.492)	(0.480)	0.288	0.280	<u>0.432</u>
Average	0.377 (0.396)	(0.385)	<u>0.530</u> (0.446)	(0.433)	0.313	0.406	0.531

Comparison with Baselines. The results in Table 2 demonstrate DSPO’s clear empirical advantage. The DSPO-trained 7B agent attains an average score of 0.531, establishing a new state-of-the-art within this comparison and corresponding to a 34.1% relative improvement over the same-sized Search-R1 (GRPO, 7B) model. More notably, the 7B DSPO agent achieves a slightly higher average score than the substantially larger Search-R1 14B models trained with either GRPO or PPO. This outcome indicates that the observed gains are attributable primarily to a more effective and stable learning algorithm, rather than to increased model scale alone.

Analysis of Ablations. The ablation results in Table 2 further show that both DSPO components are indispensable. Removing the dynamic filter (‘w/o dynamic filter’) produces a catastrophic performance decline, with the average score falling to 0.313; this result highlights the filter’s role in preventing the sequence-level objective from being starved of useful learning signal by homogeneous-reward batches. Ablating sequence-level optimization (‘w/o sequence-level opt.’) also yields substantial degradation, producing an average score of 0.406. Although this token-level variant outperforms the filter-less variant, it remains far below the full DSPO model and, as shown in the follow-

ing section, is also vulnerable to catastrophic training instability. These findings confirm that the two mechanisms are complementary: sequence-level updates are necessary for stability, whereas dynamic filtering is essential for converting sparse rewards into an efficient learning signal.

Beyond aggregate quantitative metrics, we observe that DSPO induces sophisticated search behaviors, including recognition of irrelevant results, query reformulation, and multi-turn verification (see Appendix A for detailed trajectory examples). These behaviors emerge from pure RL training under DSPO rather than from supervised demonstrations.

4.3 ANALYSIS OF TRAINING DYNAMICS

To empirically substantiate the claims concerning stability and efficiency, we analyze the training-reward dynamics of DSPO, its ablations, and key baselines. Figure 3 visualizes these dynamics and thereby provides additional evidence for the architectural choices underlying the method.

DSPO (red) exhibits a smooth, monotonic ascent and efficiently converges to the highest reward level, empirically confirming the stability conferred by the sequence-level objective. In sharp contrast, the **token-level methods**—DSPO w/o Seq-level Opt. (green) and vanilla GRPO (blue)—undergo catastrophic policy collapse early in training: after a brief initial improvement, their rewards decline precipitously, illustrating the instability induced by high-variance token-level gradient updates. Meanwhile, **DSPO w/o Dynamic Filter** (purple), which retains sequence-level updates but lacks an efficient learning signal, remains stable yet plateaus at a markedly suboptimal performance ceiling. These dynamics show that DSPO’s robust policy optimization depends on the combination of sequence-level stability and dynamic filtering.

To verify that training-reward improvements correspond to genuine generalization rather than reward hacking, we track validation performance on key benchmarks throughout training. As shown in Figure 2, DSPO’s validation accuracy on NQ, HotpotQA, and other diverse benchmarks increases consistently in a pattern that mirrors the stable reward curve. This correspondence indicates that the agent is learning a generalizable search-and-reasoning policy.

4.4 SCALABILITY AND GENERALIZATION ANALYSIS

To further assess the robustness of the approach, we extend the evaluation along two axes: model-scale transfer and domain generalization.

Scalability to Larger Models. We examine whether DSPO’s stability benefits persist at larger parameter scales by training Qwen2.5-14B-Instruct. As reported in Table 3, DSPO exhibits strong scalability: the DSPO-trained 14B model reaches an average accuracy of 60.6%, substantially outperforming the strong GRPO-14B baseline (53.0%) by a relative margin of **14.3%**. These results indicate that the method can effectively exploit increased model capacity while maintaining performance that consistently exceeds standard baselines.

Table 3: Scalability analysis for Qwen2.5-14B-Instruct, showing the effect of applying DSPO at a larger parameter scale. Best results are in **bold**.

Dataset	Instruct (14B)	GRPO (14B)	DSPO (7B)	DSPO (14B)	Gain
NQ	0.345	0.535	0.580	0.629	+17.6%
HotpotQA	0.407	0.563	0.613	0.665	+18.1%
2WikiMQA	0.332	0.611	0.569	0.699	+14.4%
Bamboogle	0.328	0.504	0.432	0.544	+7.9%
PopQA	0.364	0.477	0.498	0.545	+14.3%
TriviaQA	0.643	0.760	0.754	0.802	+5.5%
Musique	0.151	0.260	0.270	0.361	+38.8%
Average	0.367	0.530	0.531	0.606	+14.3%

Generalization to Mathematical Reasoning. We further evaluate the generality of DSPO by applying it to single-turn mathematical reasoning tasks with models from the Qwen2.5 and Qwen3 families. Table 4 reports comparisons on Math500 and Olympiad-Bench, where DSPO consistently surpasses GRPO for both 7B and 4B model sizes. These results indicate that DSPO can be effective beyond agentic search, extending to broader reasoning domains.

Table 4: Generalization to mathematical reasoning on Math500 and Olympiad-Bench. Best results are in **bold**.

Model	Benchmark	Steps	GRPO	DSPO	Gain
Qwen2.5-Math-7B	Math500	200	0.772	0.798	+2.6%
Qwen3-4B	Olympiad-Bench	100	0.728	0.755	+2.7%

5 CONCLUSION

This work addresses the critical instability and sample-inefficiency problems that arise when RL is used to train autonomous LLM search agents. We introduced Dynamic-filter Sequence-level Policy Optimization (DSPO), an algorithm that promotes robust training through two mutually reinforcing components: sequence-level optimization, which mitigates catastrophic policy collapse, and a dynamic outcome-based filter, which converts sparse rewards into a consistently informative learning signal. The experiments show that DSPO not only achieves strong performance across a suite of challenging question-answering benchmarks but also exhibits substantially greater training stability than prior methods.

By enabling robust training from environmental feedback alone, DSPO provides a practical and efficient blueprint for constructing capable LLM agents without costly expert demonstrations. On this stable foundation, future work can investigate the integration of more advanced retrievers or the extension of DSPO to complex multi-tool tasks. Because sparse rewards and unstable policy gradients are not unique to search, we further hypothesize that the same principles may yield analogous performance and stability gains in domains such as mathematics and code generation, which remain promising targets for future validation. More broadly, the core tenets of DSPO—matching the optimization unit to the reward signal and guaranteeing sufficient signal density—may be instrumental in the development of the next generation of autonomous AI systems.

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A APPENDIX

To illustrate the search strategies acquired by the model, Table A.1 presents representative trajectories. The examples show that the model can reformulate queries when initial searches return irrelevant evidence, verify information through follow-up searches, and extract comprehensive answers from retrieved results.

Table A.1: Representative trajectories illustrating the model’s learned search-and-reasoning capabilities. The examples exhibit sophisticated behaviors, including (a) recognition of irrelevant results, (b) query reformulation, and (c) verification strategies.

Example 1: Presidential Succession (<i>Query Refinement after Failure</i>)	
Question	Who is next in line if something happens to the president?
Gold	vice president
Turn 1	<code>search("who is next in line if something happens to the president")</code> → Retrieved: Documents about novels "Something Happened" and "Mosquitoes Buzz" Agent: "The queries were not directly about the line of succession... The relevant information I need is not present in the search results. Let's try searching again with a more specific query."
Turn 2	<code>search("line of succession for the president")</code> → Retrieved: Philippine and US presidential succession information Agent: "Based on the search results... the next person in line would be the Vice President."
Answer	✓ The Vice President
Example 2: University Identification (<i>Information Verification</i>)	
Question	Which American University is situated in Ithaca, New York?
Gold	Cornell University
Turn 1	<code>search(["American University in Ithaca", "University in Ithaca NY"])</code> → Retrieved: "Ithaca is home to Cornell University, an Ivy League school..." Agent: "Ithaca, New York is home to Cornell University, which is an Ivy League school. However... it is highly likely that the university in question is Cornell. Let me confirm this."
Turn 2	<code>search(["Cornell University location", "Cornell University New York"])</code> → Retrieved: Cornell history confirming establishment in Ithaca Agent: "Cornell University is indeed located in Ithaca, New York."
Answer	✓ Cornell University
Example 3: Biographical Information (<i>Direct Extraction</i>)	
Question	What is John Andrew Martin’s occupation?
Gold	lawyer, attorney, politician
Turn 1	<code>search(["John Andrew Martin occupation", "John Andrew Martin job"])</code> → Retrieved: "American journalist, attorney, soldier, and politician..." Agent: "John Andrew Martin was an American <u>journalist, attorney, soldier, and politician</u>. He represented Colorado in the U.S. House of Representatives."
Answer	✓ Journalist, attorney, soldier, and politician